v158TH FIGHTER WING
HOME OF THE GREEN MOUNTAIN BOYS
THE GREEN MOUNTAIN BOYS
HQ VERMONT AIR NATIONAL GUARD

STRIKE!

Basic Fighter Maneuvering

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Basic Fighter Maneuvering (BFM)

Basic Tenets of ACM

Outlined here are the common tenets of BFM which represent the fundamentals of air combat.

- **Lose Sight, Lose the Fight**
  - You must gain and maintain a visual with the bandit throughout the engagement.

- **Maneuver in Relation to the Bandit**
  - Observe, predict, and maneuver. Fly the aircraft to take advantage of errors committed by the bandit.

- **Nose Position vs. Energy**
  - Assess your options whether to sacrifice energy advantage for WEZ advantage, and vice versa.

- **Lead vs. Pure vs. Lag Pursuit**

Offensive BFM (OBFM)

Offensive BFM

This section will describe offensive BFM from three ranges: short, medium, and long. The objectives of offensive BFM consist of maneuvering the Weapons Engagement Zone (WEZ) and secure a kill on the bandit, and to maintain your offensive advantage.

Consider the following when performing Offensive BFM:

- Stay within the bandit's **turn circle**.
- Maneuver and manipulate the bandit's **Control Zone (CZ)**.
- Secure a **kill** with weapons employment.
- **Maintain sufficient energy** for future maneuvering.
- Maintain **mutual support** of the flight.

Offensive BFM

- Energy States
  - Typical “Fights-On” parameters in training exercises and examples consist of the following:
    - Co-Altitude
    - 40 degree Aspect Angle (AA)
    - Pure Pursuit
    - 400 KIAS ±20
  - Altitude = Potential Speed Energy
  - Speed = Potential Altitude Energy

Short OBFM

If within 3,000 feet of the bandit, you should be on or in the turn circle, and close if not within the CZ. This can occur through the adversary's lack of awareness, or the result of effective OBFM. Your goal during short OBFM is to remain in the CZ, maintain closure, and

- CZ:
  - Once established in the Control Zone (CZ), closely monitor your Aspect Angle (AA) when maneuvering for weapons employment. Do not pass up or make unnecessary sacrifices of speed or altitude energy for turn rate.

- Transitions:
  - While following the adversary through a maneuver, it is important to be aware of the amount of turning room between your aircraft and the floor. By flying a shallower vertical flight path, altitude and speed energy will be preserved.
  - ROT: Dive attitude should not exceed twice the altitude to the floor.

Short OBFM

- Overshoots:
  - The attacking aircraft has numerous problems to solve in pursuit of the adversary aircraft. Two of the biggest problems, excessive AA and excessive closure, **will cause an overshoot** if not corrected. **There are three types of overshoots:**
    - **Flight Path**: Occurs when the attacker's aircraft crosses to the outside of a defender's turn circle.
    - **Longitudinal**: Occurs when the attacker crosses the longitudinal axis of the defender.
    - **3/9 Line**: Occurs when the attacker crosses the lateral axis of the defender.

Short OBFM

- Overshoots (cont.):

  - **Quarter Plane:**
    - The Quarter Plane is an aggressive maneuver to prevent an FP or 3/9 line overshoot. Execution consists of rolling wings level, pulling max G until a suitable closure rate is achieved, and beginning a lead pursuit curve to settle into the CZ.

  - **Lag Roll:**
    - The Lag Roll is a maneuver that controls closure and prevents a longitudinal overshoot by rolling around the flight path of the defender and increasing the distance flown in relation to actual ground distance. It will decrease AA at the expense of aspect. Execution consists of pulling a moderate amount of G and rolling the opposite the direction of the turn. This maneuver should be used only in lead pursuit.

Short OBFM

- Overshoots (cont.):
  - **Scissors:**
    - If an attacking aircraft overshoots, it's possible that the defending aircraft will reverse the turn. If this happens, the attacker must aggressively maneuver and pull to the defending aircraft's high six. If the defender counters with the same maneuver, it is likely that a scissor will develop. There are three types: flat, rolling, and vertical.
      - Flat Scissors:
        - Occurs when the overshoot is in the turning plane and both fighters are co-altitude.
      - Rolling Scissors:
        - Occurs when the overshoot is off the turning plane.
      - Vertical Scissors:
        - Occurs when the overshoot is in the vertical plane.

Medium OBFM

Medium-Range Offensive BFM occurs within 6,000 feet of the adversary, and teaches how to enter an adversary's turn circle, maneuver to the CZ, and fine tune your position for weapons employment.

- Entry Window:
  - Fly into the adversary's turn circle by rolling out and flying to a point approximately one-third in from where the defender started the break turn and the center of the turn circle.

- Maneuvering to the Turn Circle:
  - Decrease load factor, and accelerate toward the adversary. Getting to the turn circle in a short time decreases the adversary's ability to increase AA.

- In the Turn Circle:
  - Utilize pursuit curves and Yo-Yos to maneuver into the CZ and transition to short BFM.

Medium OBFM

- Yo-Yo's:
  - High Yo-Yo:
    - A High Yo-Yo is a lag pursuit curve executed out of the defender's turning plane, and is initiated when the attacker places the aircraft in lag pursuit, increasing load on the aircraft above the defender's turning plane, and maneuvering back in. The result is a solution to angle, closure, or range options that can't happen in-plane.
  - Low Yo-Yo:
    - A Low Yo-Yo is a lead pursuit curve executed inside the defender's turning plane, and is initiated when the attacker places the aircraft in lead pursuit, maneuvering the aircraft below the defender's turning plane, and maneuvering back in. The result is similar to a High Yo-Yo.

Long OBFM

Long-Range Offensive BFM occurs within 9,000 feet of the adversary aircraft, beginning with a 3/9 line advantage at a range outside of the adversary's turn circle. With the range, the adversary can easily pose geometry against the attacking aircraft, creating a neutral fight at the merge.

- Snapshot:
  - The attacking aircraft may be driven to attempt a high-aspect gunshot or missile shot. Regardless of the circumstances, the attacking aircraft must be within range and plane of motion with adequate lead pursuit on the adversary.

- Entering the Turn Circle:
  - If a snapshot fails or is not taken, maneuver into the adversary's turn circle and CZ. Do not pull into constant lead or pure pursuit, as such will waste speed energy.

Defensive BFM
(DBFM)

Defensive BFM

This section will describe three primary objectives in Defensive Basic Fighter Maneuvering:

- **Defeat**
    - Outperform the adversary with potential energy and turn rate.
- **Deny**
    - Avoid acquisition or a weapons solution by the adversary.
- **Neutralize**
    - Force an effective overshoot or a successful reversal.

Defensive BFM

- Adversary Acquisition:
  - If the defender acquires the adversary outside of the turn circle, the defender should assume the adversary has visual contact, and has a positive rate of closure.
  - If the adversary pulls into any pursuit, continue with a maximum performance, high load factor turn. Turning angle prior to the adversary pulling pure or lead pursuit is more geometry for the adversary to solve.
  - As the adversary enters the defender's turn circle, they will attempt to solve closure and geometry with in and out of plane maneuvers. The defender's goal is to defeat the adversary's desired weapon solution.

Defensive BFM

- Initial Moves:
  - Once the adversary is acquired, react. The appropriate reaction depends on performance, geometry, and range. The initial move creates geometry problems for the adversary. There are three types of defensive turns:
    - Navigation: An energy sustaining turn at low load factor to assess the adversary's intentions.
    - Hard: An energy sustaining turn at moderate load factor to deny a weapons solution.
    - Break: A maximum performance turn at high load factor to defeat a weapons solution or force an overshoot.
- Defensive Cues:
  - Accurate assessment and understanding of fundamentals are the key to the success in BFM engagements. To counter an attack, Observe, Predict, Maneuver, and Assess.

Defensive BFM

- Overshoot:
  - There are two types of overshoots to take advantage of in any engagement:
    - A longitudinal overshoot, wherein the adversary crosses the tail-nose line of the defender.
    - A lateral overshoot, wherein the adversary crosses the 3/9 line of the defender. Typically seen as the traditional type of overshoot in an ACM scenario.
- Reversal:
  - Following a longitudinal overshoot, though optimally in anticipation of a longitudinal overshoot, the defender should reverse their plane of motion, placing their lift vector on the adversary as they overshoot, and forcing the fight into a neutral stance, likely in the form of flat scissors.

High Aspect BFM (HABFM)

HABFM

High Aspect Basic Fighter Maneuvering (HABFM) can occur during close-range phases of any tactical engagement. While ideally prevented through a lead turn, or denied through BVR tactics altogether, it's critical to know which moves to play. Remember, in a neutral fight, any turning room for you is turning room for the adversary. There are four primary concepts to understand during neutral/HABFM:

- Merge: Both aircraft cross each other's 3/9 line, wherein, given no lead turn has been performed, the fight is neutral, and the aircraft are tail-to-tail.
- Lead Turns: Either aircraft turns early to enter a lower aspect on the other aircraft prior to a merge.
- One Circle: Occurs when both aircraft turn in the same direction post-merge, inscribing a flight path of a single circle over the ground. Preferred in jets with a smaller turning radius than the adversary.
- Two Circle: Occurs when both aircraft turn in opposite directions post-merge, inscribing a flight path of a single circle over the ground. Preferred in jets with a higher sustained turn performance.

End

You have completed the lesson portion covering BFM.